Department of Electrical Engineering,
Faculty of Engineering and Technology,
Jamia Millia Islamia, New Delhi

Academic Session -2026-2027
1st Mid Semester Examination B Tech. (Electrical & Computer Engineering)

SUBJECT: - EEC-509 Object Oriented Programming

TIME:01 Hours **Max.Marks:30**

Note:

1. *M-Marks, L-Bloom's Level,C–Course Outcome*
2. *Electronic gadgets are not allowed in the examination hall like mobile phone, smart watch etc.*
3. *Students will only be allowed to use non-programmable scientific calculator.*

			M	L	COs
Q1	**All questions are compulsory**				
	a	Define a class and an object in C++.	01	L1	CO1
	b	List any two differences between structured programming and object-oriented programming.	01	L1	CO1
	c	What is compile-time polymorphism?	01	L1	CO2
	d	What is method overriding?	01	L1	CO2
	e	Define a pure virtual function.	01	L1	CO3
	f	What is hybrid inheritance?	01	L1	CO3
Q2	**All questions are compulsory**				
	a	Compare and contrast the default constructor, parameterized constructor, and copy constructor in C++ with suitable examples.	06	L2	CO1
	b	Explain the concept of encapsulation in C++ with a suitable example. How is data hiding achieved using access specifiers (private, public, protected)? Also discuss the role of getter and setter methods.	06	L2	CO1
Q3	**All questions are compulsory**				
	a	Analyse how abstraction is achieved in Java using abstract classes and interfaces. Critically differentiate between abstract class and interface on the basis of their features, limitations, and design implications, with suitable examples.	06	L3	CO2
	b	Apply the concepts of C++ templates and Java generics to demonstrate their working. Illustrate the differences in compile-time behaviour, type safety, and implementation with suitable programming examples.	06	L3	CO2

